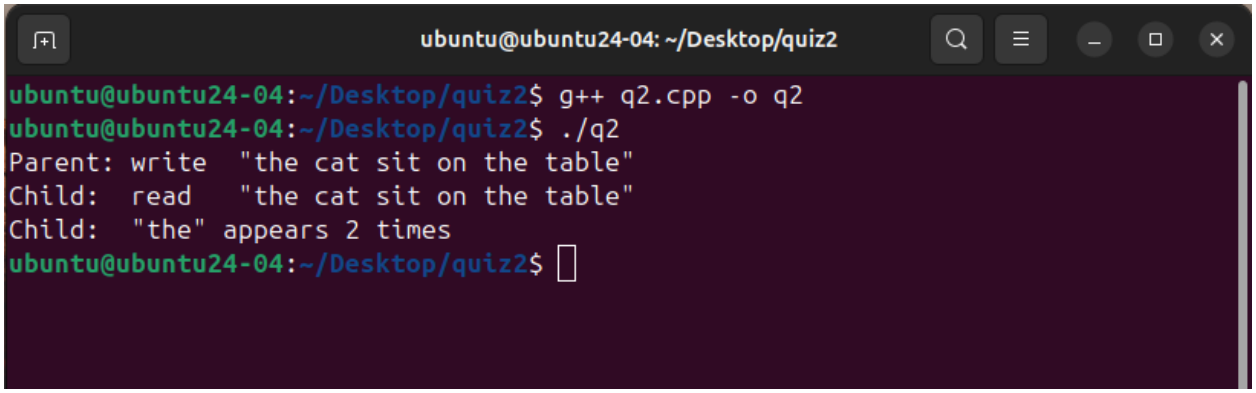
```
#include<iostream>
#include<sys/ipc.h>
#include<sys/shm.h>
#include<stdio.h>
#include<string.h>
#include<unistd.h>
#include<sys/wait.h>
using namespace std;
int countOccurrences(char *str, char *sub) {
int count = 0;
int subLen = strlen(sub);
char *pos = str;
  while ((pos = strstr(pos, sub)) != NULL) {
    count++;
    pos += subLen;
  }
  return count;
}
int main() {
key_t key = 1234;
char sentence[] = "the cat sit on the table";
char substring[] = "the";
int shmid = shmget(key, 256, IPC_CREAT | 0666);
    pid_t pid = fork();
if (pid > 0) {// parent
  char *shm = (char *)shmat(shmid, NULL, 0);//attach
  strncpy(shm, sentence, 255);
  printf("Parent: write  \"%s\"\n", shm);
  shmdt(shm);
  wait(NULL);
  shmctl(shmid, IPC_RMID, NULL);
    }
else {// child
  sleep(1);
  char *shm = (char *)shmat(shmid, NULL, SHM_RDONLY); //read
  printf("Child:  read   \"%s\"\n", shm);
  int count = countOccurrences((char *)shm, substring); //count
  printf("Child:  \"%s\" appears %d times\n", substring, count);
  shmdt(shm); //detach
    }
  return 0;
```

```
}
```


ubuntu@ubuntu24-04: ~/Desktop/quiz2
ubuntu@ubuntu24-04:~/Desktop/quiz2$ g++ q2.cpp -o q2
ubuntu@ubuntu24-04:~/Desktop/quiz2$ ./q2
Parent: write "the cat sit on the table"
Child: read   "the cat sit on the table"
Child: "the" appears 2 times
ubuntu@ubuntu24-04:~/Desktop/quiz2$